

Pretext

An AI sparring partner for frontline staff

Phishing simulations trained people to stop clicking.

Pretext trains them to stop caving on the phone.

An AssemblyAI Voice Agent plays the *caller*.

You play the employee.

A live rubric scores the call. A protected-action trip-wire decides

BREACH or **HELD THE LINE**.

The problem

- Voice social engineering (**vishing**) of help desks is behind headline breaches — the attacker just calls and *asks*.
- Security-awareness vendors sell phishing sims by the seat; **human-run vishing sims are expensive and rare** — most teams never drill the phone.
- Contact centers pay for onboarding role-play that's scripted, static, and can't escalate when the trainee freezes.

The idea

Make the *attacker* a voice agent: adaptive, adversarial, always available — with a deterministic trip-wire that decides the moment the trainee gives away the protected action.

Demo flow

Four personas at fictional **Northwind Utilities**:

Scenario	Caller	Protected action
The CFO Pretext (flagship)	"Dana Whitfield" — locked out before a board call	MFA / password reset
The Double Charge	"Ray Delgado" — furious, threatens to cancel	Refund > \$50
Rosa's Letter	"Rosa Vega" — 74, ES→EN, digit-by-digit	Account details
The Remittance Switch	"Mark Ellison" — vendor BEC	Payment-details change

Beat: the moment the trainee caves, the agent calls

`attempt_protected_action` → **BREACH** banner → the same voice drops character and coaches.

How it works

browser (Next.js)

AssemblyAI

```
mic → AudioWorklet PCM16@24kHz —WS—▶ Voice Agent (STT+LLM+TTS)
    ◀— reply.audio / transcripts / tool.call —
lib/reducer → lib/rubric + lib/metrics → verdict + debrief
director: conversation.message nudges · session.update escalation
        reply.create final push · coach switch
```

- **Token route** mints capped tokens (≤ 240 s, per-IP rate limit, daily cap, required private passcode) — the API key never reaches the browser.
- **Mock agent + replay fixtures** — offline dev, CI, and a labeled fallback demo path.

AssemblyAI feature map

Feature	How Pretext uses it
Voice Agent WS	The entire call: STT, turn-taking, persona, TTS, barge-in
<code>GET /v1/token caps</code>	Hard ≤ 240 s sessions on a public demo
Stored agents	Persona prompts + tool schemas server-side
Client-side tools	<code>attempt_protected_action</code> = trip-wire; <code>log_observation</code> = rubric
<code>session.update</code>	Escalation + persona→coach switch
<code>conversation.message</code> / <code>reply.create</code>	Director channel
<code>transcript.*</code> / <code>input.speech.*</code>	Captions, rubric regexes, latency/interruption metrics
<code>input.keyterms</code> / <code>transcription_prompt</code>	Fictional names, digits, ticket formats
<code>input.turn_detection</code>	Barge-in vs patient pacing per persona
Session history API	Recording + timeline on the debrief page

Honest differentiation

- **Adversarial, adaptive AI caller** with a protected-action trip-wire — not a chatbot, not a detector.
- **Evidence-backed verdicts** — every rubric hit points at the transcript span, tool call, or metric that triggered it; labeled mock replays are available now, and live browser recording playback awaits verification.
- **Deterministic where it counts** — latency, talk ratio, interruptions, trip-wire, policy-phrases regexes. Soft-skill judgments are agent-observed and labeled as such.
- **Everything on one API** — no separate STT/TTS/LLM vendors; the trainee uses a browser while the server keeps the API key and stored-agent IDs private.

Verification status

Mock-verified: 8 Playwright E2E and 41 unit tests cover the full call, trip-wire verdict, coach debrief, and JSON export. Lint, typecheck, and build pass.

Production integration smoke test passed: `/api/status` returned `live_ready: true`, `/api/token` returned HTTP 200, four stored agents were configured, and a genuine AssemblyAI session produced a greeting, coach response, and retrievable history.

Still to verify: the full deployed browser trainee flow, call quality and verdicts across all four personas, and live browser recording playback.

Demo integrity: all eight replays and bundled videos are labeled recorded mock. Replace the footage after recording a genuine browser call.

Live AssemblyAI verification progress

- [x] Publish and configure four stored agents; production status ready.
- [x] Mint a live token and verify a genuine WebSocket greeting, coach response, and session-history fetch.
- [] 3 live calls per scenario (12 total, ≤ 240 s each): confirm persona holds character, trip-wire fires correctly, coach switch works.
- [] Capture per-call numbers: greeting TTFB, reply latency p50/p95, turn-detection accuracy, STT accuracy on names/digits (keyterms on).
- [] Verify a full deployed browser call and recording playback in debrief.
- [] Record scripted live demo (video_script.md) replacing demo.mp4.
- [] Finalize and submit the draft LabLab entry.

The current demo video and GIF are mock/replay captures. LabLab is draft.

Business model & market

- **Buyers:** CISO / security-awareness lead (vishing drills), contact-center training lead (de-escalation & verification reps).
- **Pricing story:** per-seat per-month + scenario packs (help-desk pretext pack, billing-pack, BEC pack); enterprise: custom personas + rubrics wired to internal policy.
- **Why now:** voice agents just got good enough to be a believable attacker — and help-desk vishing keeps making headlines.

Limitations (said out loud)

Persona drift · trip-wire false $\pm$ · STT digit errors · echo on Firefox/Safari without headphones · API outage (Replay mode is the fallback) · scoring is a training aid, not certification · single-mic, no diarization · personas target only the fictional Northwind Utilities — defensive training, same category as phishing sims.

Roadmap

1. Real-time persona packs & rubric authoring UI
2. Team dashboards: cohort scores, trend lines, recurring drills
3. More personas: IT-recruiter pretext, "auditor", insider-threat calls
4. Deeper session-history analytics (timeline-aligned rubric events)
5. Optional LLM-judged soft-skill scorecard (when funded)

Pretext — train the call before it costs you.